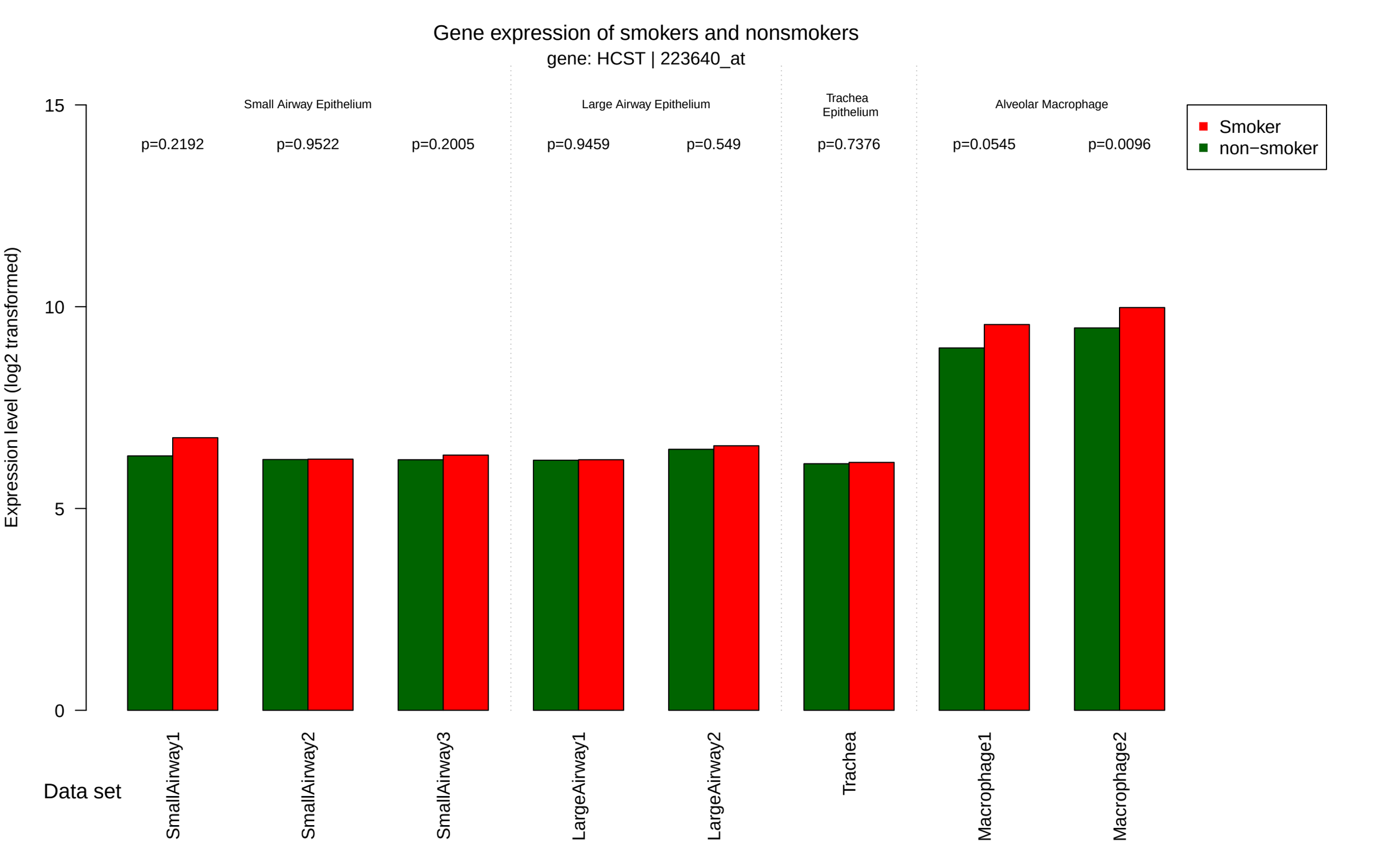
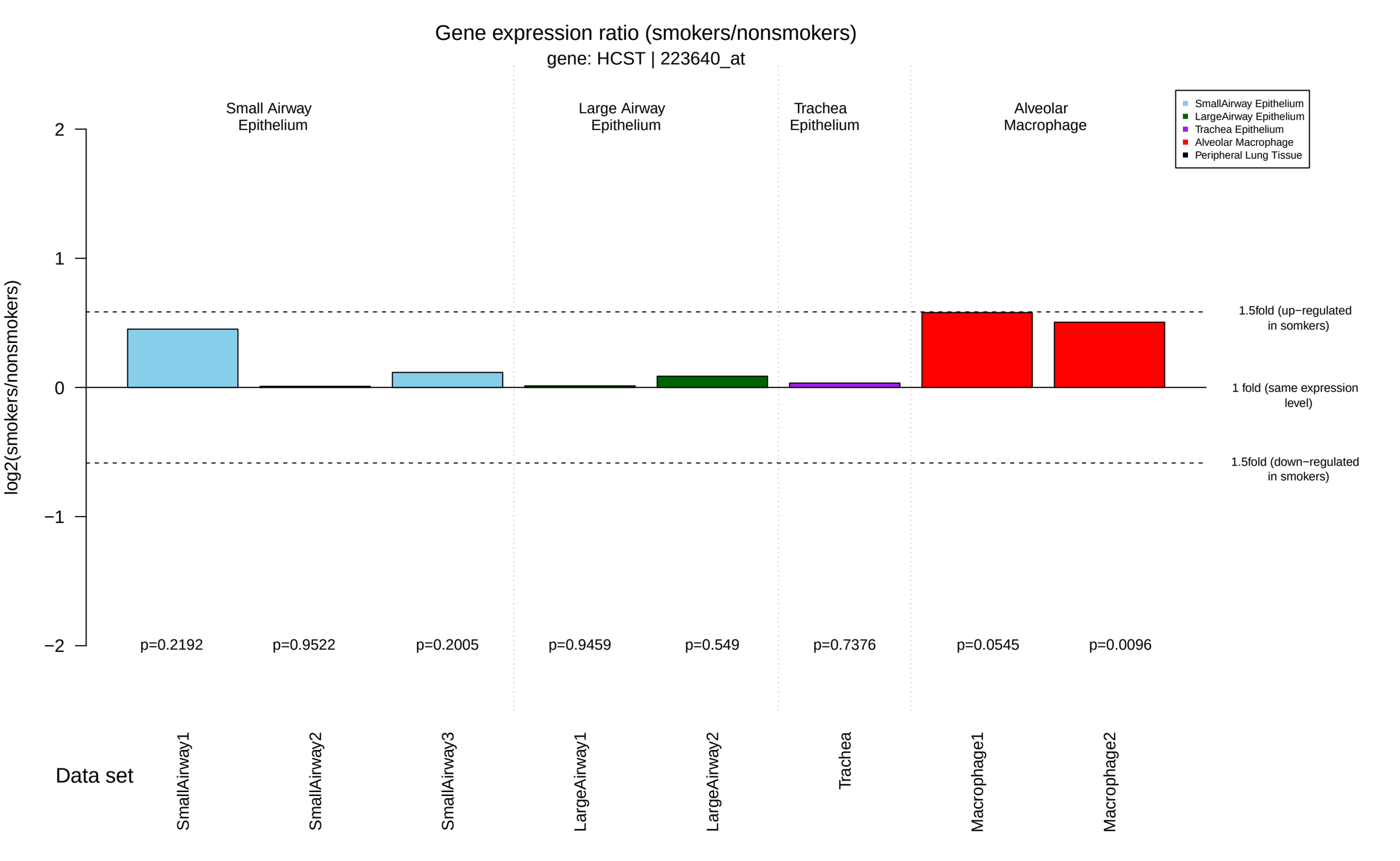
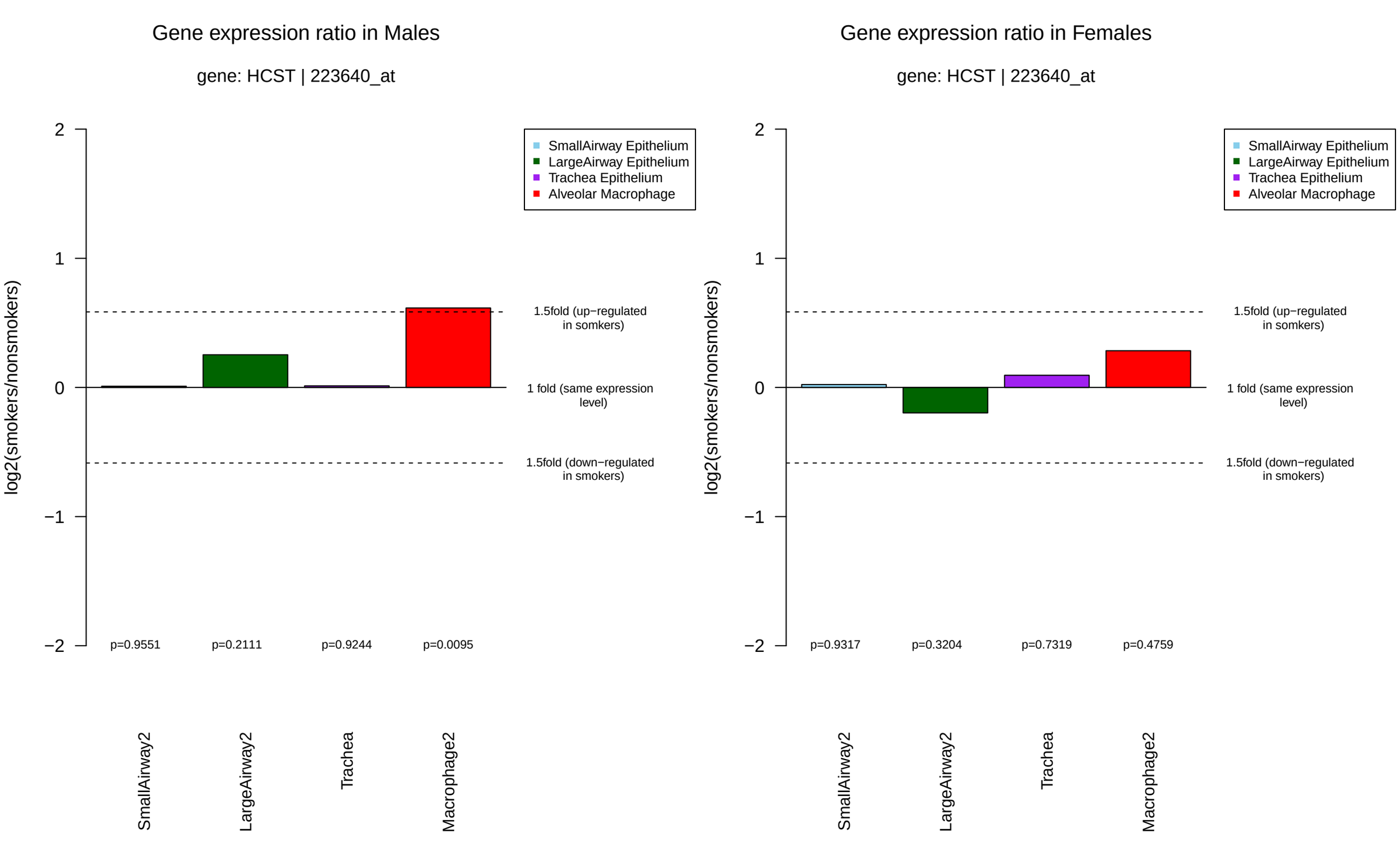
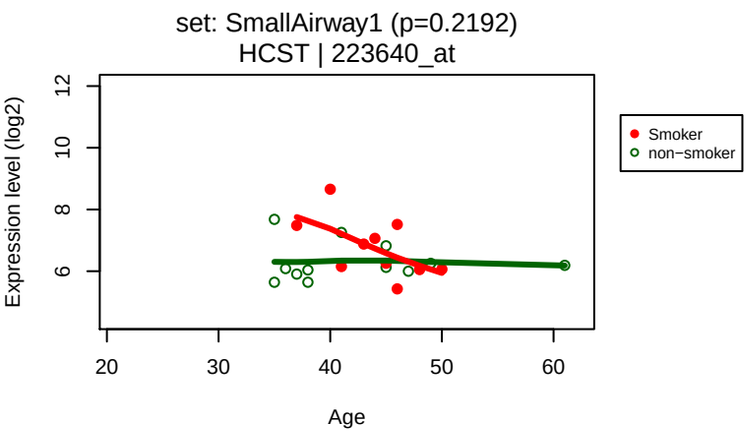
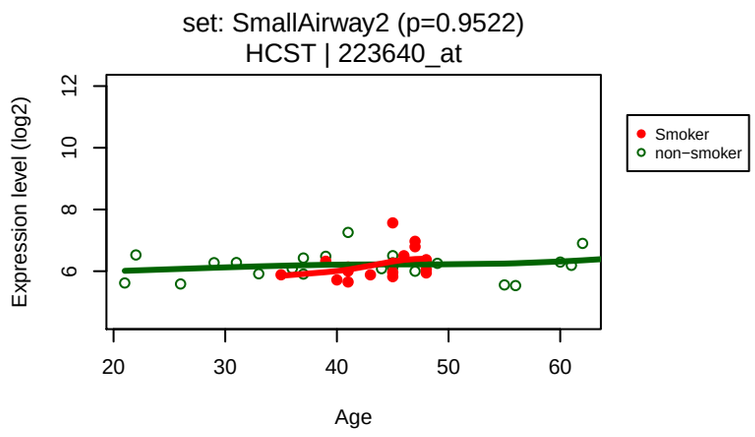
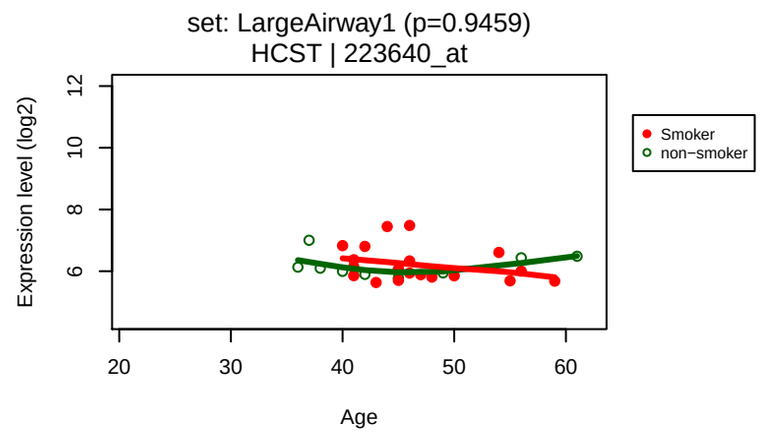

Gene expression of smokers and nonsmokers
gene: HCST | 223640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2192
p=0.9522
p=0.2005
p=0.9459
p=0.549
p=0.7376
p=0.0545
p=0.0096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

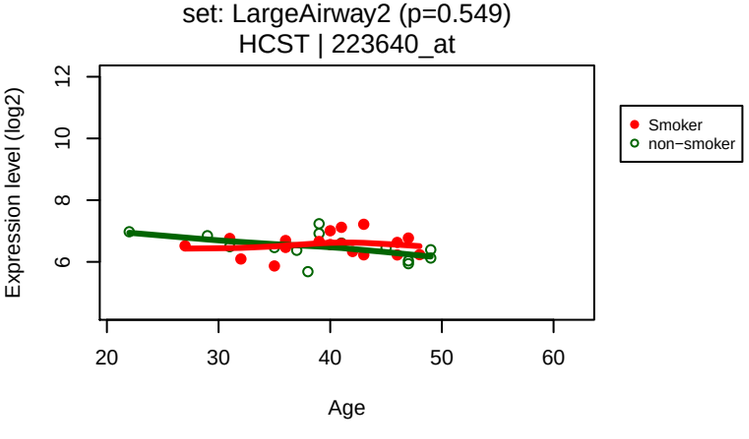

Gene expression ratio (smokers/nonsmokers)
gene: HCST | 223640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2192
p=0.9522
p=0.2005
p=0.9459
p=0.549
p=0.7376
p=0.0545
p=0.0096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

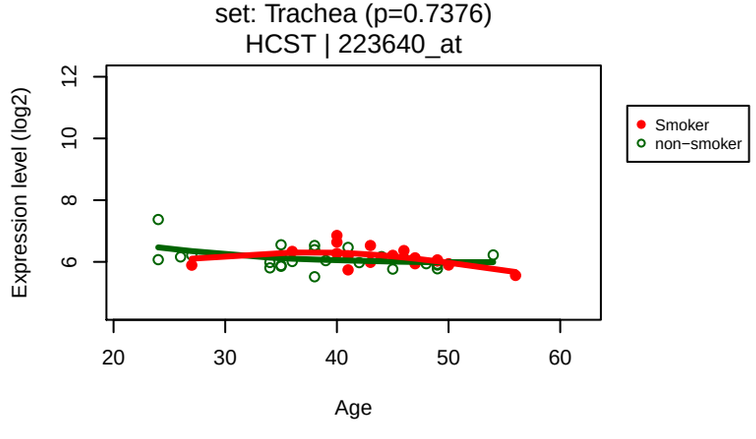

Gene expression ratio in Males
gene: HCST | 223640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9551
p=0.2111
p=0.9244
p=0.0095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCST | 223640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9317
p=0.3204
p=0.7319
p=0.4759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

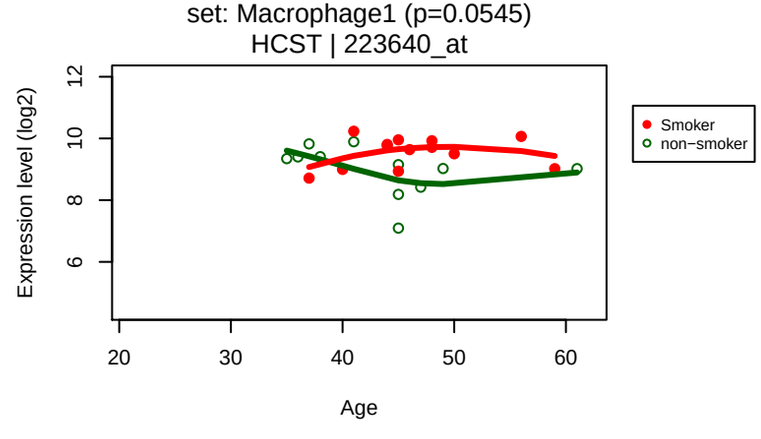

set: SmallAirway1 (p=0.2192)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker

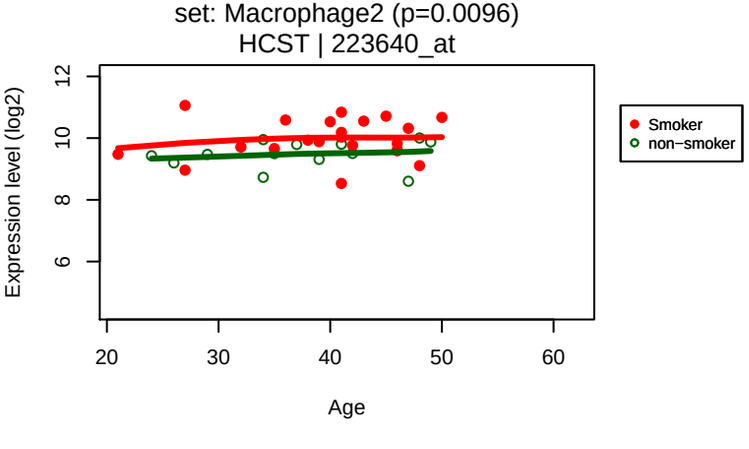

set: SmallAirway2 (p=0.9522)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker

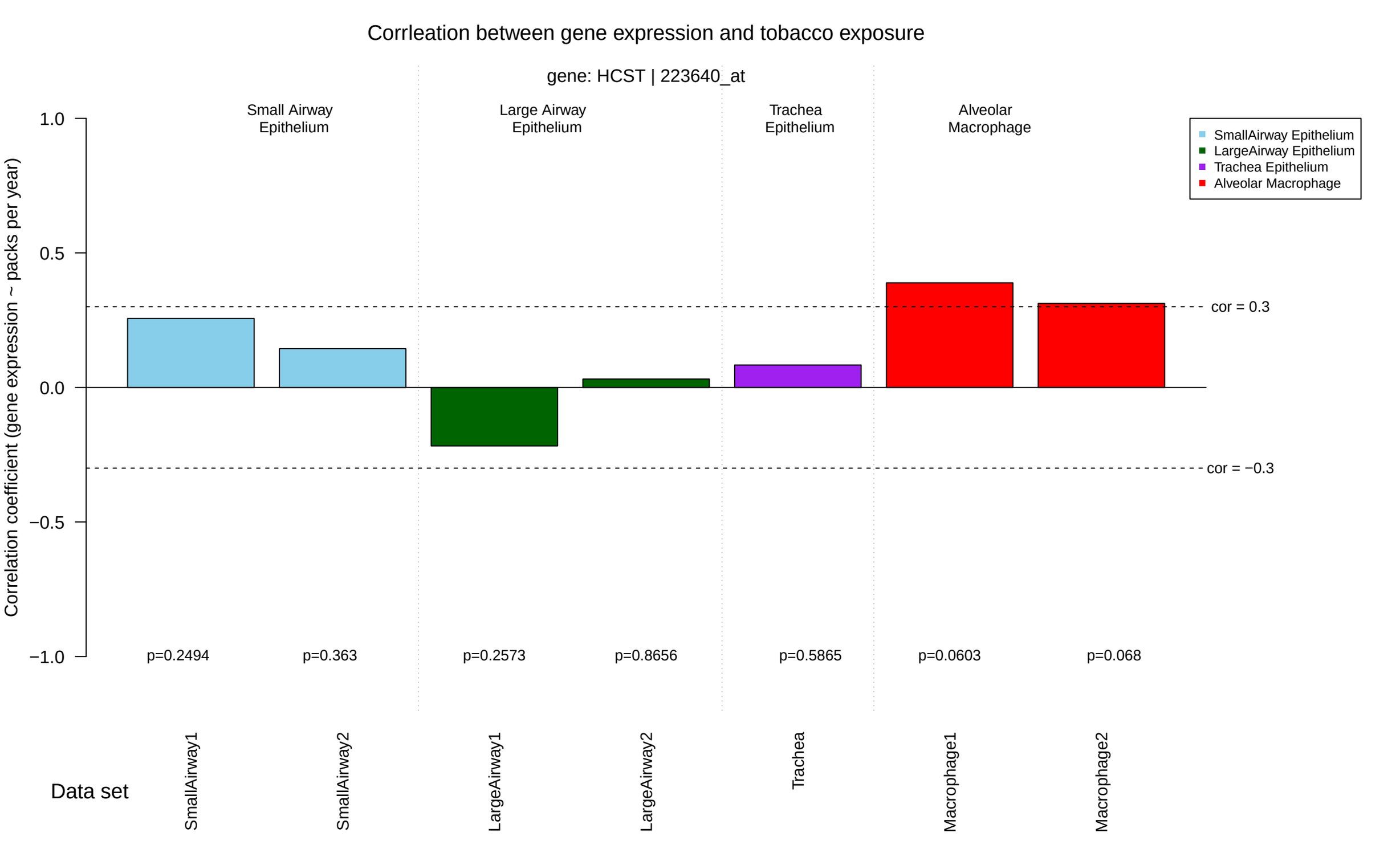

set: LargeAirway1 (p=0.9459)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.549)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7376)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0545)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0096)
HCST | 223640_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCST | 223640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2494
p=0.363
p=0.2573
p=0.8656
p=0.5865
p=0.0603
p=0.068
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2